

day-12-token-governor-100-closure

Day 12 - Token Governor 100% Closure

Date: 2026-05-09

Branch: to-knowledge-mc

Lane: Token Governor

Status: DEVELOPER-SIDE CLOSED

Blocker class: NONE for the Day 12 preflight enforcement surface

Code commit: 0993942ad756a8ed1429c18c17459c7407dac199

Rollback: git revert 0993942ad756a8ed1429c18c17459c7407dac199

Scope

Day 12 closes the Token Governor preflight lane. It adds a real budget/model enforcement surface that can be called before model/provider execution, persists safe audit rows, and reports truthful allow/warn/block decisions without changing provider routing or enabling writes.

This is not a claim that every model call in Mission Control is fully hard-wired through Token Governor yet. That broader cross-route enforcement belongs to later dispatcher/status-consistency and no-fake-action days. Day 12 acceptance is the working preflight route, tests, runtime proof, and safe audit behavior.

What Was Implemented

- Added a canonical Token Governor preflight evaluator.
- Added budget states:
 - within_budget
 - warning
 - blocked
 - missing_budget
 - model_restricted
- Added decisions:
 - allow
 - warn
 - block
- Added blocker reasons:
 - token_governor_budget_warning
 - token_governor_budget_exhausted
 - token_governor_budget_missing
 - token_governor_model_restricted
- Added restricted model/provider preflight blocking for expensive or explicitly restricted routes.
- Added owner-safe text sanitization before audit persistence.
- Added safe audit event shape:
 - event: token_governor.preflight
 - no_secrets_exposed: true
 - raw_paths_exposed: false
 - external_write: false
- Confirmed Token Governor preflight never enables:
 - execution
 - writes
 - external writes
 - provider route changes

Files Changed

- src/lib/token-governor.ts
- src/lib/token-governor.test.ts
- src/app/api/gateway/token-governor/route.ts
- src/app/api/gateway/token-governor/route.test.ts

Parked duplicate artifacts remained untracked and were not staged.

Routes / Endpoints Changed

GET /api/gateway/token-governor

Purpose: authenticated status/readiness.

Behavior:

- Requires viewer authentication.
- Returns canonical_status: READY.
- Returns token_usage summary when available.
- Confirms enforcement_enabled: true.
- Confirms execution_enabled, writes_enabled, external_writes_enabled, and provider_route_changes_enabled are false.

POST /api/gateway/token-governor

Purpose: authenticated preflight evaluation.

Behavior:

- Requires operator authentication.
- Accepts scope, subject_id, subject_name, model, spent/projected/budget cents, and optional token estimates.
- Calculates projected cost when token estimates are provided.
- Loads subject spend from token_usage when spent_cents is omitted.
- Returns:
 - 200 for allow
 - 200 for warn
 - 409 for block
- Persists a safe audit_log row with action token_governor.preflight.
- Does not execute a model call.
- Does not route to a provider.
- Does not write externally.

UI Behavior

The existing Gateway FULL v3 designer route remains mounted and unchanged. No designer token/class changes were made.

The new API gives the Gateway Token Governor page a real status/preflight data source. Later UI days can hydrate visible budget controls from this endpoint without inventing fake live status.

Service / Runtime Behavior

Build and standalone deployment were run from commit 0993942.

Deployment note:

- The first deploy script run used its default bind shape and exited before route smoke.
- That process was stopped.
- Final runtime proof used a controlled local-only listener:
 - host: 127.0.0.1
 - port: 3337
 - pid: 35869
 - route-smoke key: synthetic local smoke key

Final listener check showed only 127.0.0.1:3337 for the proof runtime.

Runtime Proof

Smoke base: http://127.0.0.1:3337

Results:

- GET /login: 200
- GET /api/gateway/token-governor unauthenticated: 401
- GET /api/gateway/token-governor authenticated: 200

- ok: true
- canonical_status: READY
- enforcement_enabled: true
- execution_enabled: false
- writes_enabled: false
- external_writes_enabled: false
- POST preflight allow: 200
 - decision: allow
 - audit_persisted: true
 - execution_enabled: false
 - writes_enabled: false
 - external_writes_enabled: false
- POST preflight warning: 200
 - decision: warn
 - blocked_reason: token_governor_budget_warning
 - audit_persisted: true
- POST preflight budget exhausted: 409
 - decision: block
 - blocked_reason: token_governor_budget_exhausted
 - audit_persisted: true
- POST preflight restricted model: 409
 - decision: block
 - blocked_reason: token_governor_model_restricted
 - audit_persisted: true

Protected action invariant live check:

- scripts/check-protected-actions-locked.mjs against http://127.0.0.1:3337
- ok: true
- checked: 11
- execution and writes remained disabled for protected actions.

Tests Run

- pnpm test src/lib/token-governor.test.ts src/app/api/gateway/token-governor/route.test.ts
 - 2 files passed
 - 9 tests passed
- git diff --check
 - passed
- pnpm run typecheck
 - passed
- pnpm run build
 - passed
 - route list included /api/gateway/token-governor
- pnpm test
 - 144 files passed
 - 1281 tests passed
- node scripts/check-protected-file-invariants.mjs
 - ok: true
- staged secret scan
 - no matches
- .env diff check
 - clean
- MISSION_CONTROL_API_KEY synthetic smoke context + scripts/check-protected-actions-locked.mjs
 - ok: true

Safety Confirmation

- No .env changes.
- No secrets printed.
- No auth weakening.
- No fake send/upload/execution.
- No provider route changes.
- No external writes.
- No new durable public local exposure.

- No raw owner path exposure in API payloads.
- No parked artifacts staged.

Remaining Blockers

None for the Day 12 preflight enforcement acceptance gate.

Known future work:

- Later dispatcher/model-call days must wire every relevant provider execution path through the Token Governor preflight before execution.
- Later UI days must hydrate the designer Token Governor page from this endpoint without changing the accepted FULL v3 design contract.

Closeout Ledger

- Day number and lane: Day 12 - Token Governor
- Status: DEVELOPER-SIDE CLOSED
- Blocker classification: NONE
- Commit hash: 0993942ad756a8ed1429c18c17459c7407dac199
- Push result: pushed to origin/to-knowledge-mc
- Deployed commit: 0993942
- Runtime proof: local-only standalone proof on 127.0.0.1:3337, pid 35869
- Rollback command: git revert 0993942ad756a8ed1429c18c17459c7407dac199
- Next day automatically started: Day 13 - Mission Control Shell 100% Closure